

Alberta Grade 3 & 4 Science

Combined-Class Co-Teaching Plan

Based on Alberta Education new K–6 Science curriculum (finalized March 2023)

Official: curriculum.learnalberta.ca · Snapshot: alberta.ca/curriculum-science

Bottom line: Both grades share the same seven organizing ideas. Teach shared big ideas together, then use grade-specific exit tickets, journals, and projects so each grade still meets its learner outcomes.

1. How the Two Grades Are Organized

Both grades use the same seven organizing ideas. That is the main reason a split 3/4 class can share many lessons.

Organizing Idea	Grade 3 Focus	Grade 4 Focus
Matter	Substances change, including water & water cycle	Waste management & environmental impacts
Energy	Contact forces & simple machines	Gravity & magnetism
Earth Systems	Surface change; layers/fossils; FNMI land knowledge	Earth systems & conservation
Living Systems	Plant/human/animal/environment interactions	External structures & sensory organs
Space	Not a Grade 3 focus	Objects in space connected to daily life
Computer Science	Creativity + computational thinking	Design processes to solve problems
Scientific Methods	Sources, accuracy & analysis of data	Role of data & evidence

2. Similarity Map — What You Can Co-Teach

Strongest co-teach overlaps

Shared Theme	Grade 3	Grade 4	Strategy
Water & Earth systems	Water cycle; states; safety	Hydrosphere; conservation	Shared experiment; split products
Stewardship / impact	Land change; farming; respect	Waste management; 6 Rs	Shared inquiry; split action plans
Forces	Contact forces; simple machines	Gravity; magnetism	Shared language; grade stations
Living systems	Food chains; human effects	Structures & senses	Shared walk; split analysis
Inquiry / design	Data accuracy; creativity	Evidence; design process	Same investigation, deeper analysis

Partial overlaps

Theme	Shared Launch	Grade-Specific Finish
Matter / materials	Properties; natural vs processed	Gr 3: changes of state/water cycle. Gr 4: hazards & waste plan
Earth surface stories	Alberta landscapes & erosion	Gr 3: fossils/FNMI land. Gr 4: spheres sustaining life
Computer Science	Steps / creative problem solving	Gr 3: divergent thinking. Gr 4: full design cycle

Teach separately

Grade 3-Only	Grade 4-Only
Simple machines design challenge	Space & daily life
Fossils / paleontology deep dive	Magnetism investigations
Contact-force + machines investigations	Formal waste / dangerous materials unit

Tip: Use workshop rotation for unique topics — one grade with you, the other on inquiry/reading/CS.

3. Suggested Year Sequence

Block	Joint Theme	Grade 3 Must-Hit	Grade 4 Must-Hit
1	Matter + Scientific Methods	Changes of state; water cycle	Waste & dangerous materials
2	Energy (Forces) + CS	Contact forces; simple machines	Gravity & magnetism
3	Earth + Living Systems	Surface change; fossils; interactions	Spheres; conservation; structures
4	Capstone + Space	Integrated inquiry fair	Space & daily life + evidence

4. Co-Teachable Unit Kits

Unit A — Water, Change, and Earth's Systems

Gr 3: substances change / water cycle · Gr 4: water in Earth systems + conservation

Activities

- Bag/jar mini water cycle — Gr 3 labels cycle stages; Gr 4 links hydrosphere & conservation.
- Alberta seasonal water walk (include ice safety).
- Classroom water/litter audit → conservation action plan.

Videos

- Crash Course Kids – The Great Aqua Adventure (#24.1)
- NASA Exploring the Water Cycle; NASA GPM Animated Water Cycle
- PBS LearningMedia – Water Cycle Animation (Clue into Climate)

Books

- The Water Cycle (Bobbie Kalman); Water (Nat Geo Kids Level 3)
- The Lost Drop; Autumn Peltier, Water Warrior

Unit B — Waste, Land Change, and Stewardship

- Waste sort (reduce/reuse/recycle/repurpose/repair/compost).
- Before & after Alberta land photo inquiry; hazard symbol hunt (Gr 4).
- Books: Little Land; Nature Is a Sculptor; The Dirt Book; EPS Waste and Our World activities.

Unit C — Forces: Machines + Gravity/Magnets

- Stations: push/pull & friction (both); levers/ramps/wheels (Gr 3); magnets & gravity (Gr 4).
- Videos: Crash Course Kids Defining Gravity; PBS Simple Machines.
- Books: Push-Pull Morning; Hands-On Science: Motion; Magnets Push, Magnets Pull.

Unit D — Living Systems: Interactions + Structures

- Schoolyard organism census; local food chains (Gr 3); structure–function cards (Gr 4); senses/light experiments.
- Books: Creep, Leap, Crunch!; Bringing Back the Wolves; Finding Moose; Raven, Rabbit, Deer; Max Axiom Ecosystems.

Unit E — Fossils & Alberta Land Stories (Gr 3 heavy)

- Cookie/layered fossil dig; Badlands virtual tour; FNMI land knowledge via school protocols.
- Books: Dinosaurs of the Alberta Badlands; The Fossil Whisperer; Kid Paleontologist.

Unit F — Space & Daily Life (Grade 4 required)

Run as a Grade 4 focus block; Grade 3 on inquiry/CS centres. Hub: ULethbridge Gr 4 Space + APLC Space PD.

5. Best Alberta One-Stop Resource Hubs

- Official outcomes — curriculum.learnalberta.ca (Science)
- LearnAlberta resources + Pearson Spark eBooks (teacher login)
- ULethbridge Curriculum Lab — library.ulethbridge.ca/Science3 and Sciencegrade4
- Science Curriculum Wayfinder — sciencecurriculumwayfinder.ca
- APLC CPAR — apl.c.ca (sample activities & assessments)
- CESD New Curriculum site; RDPSPD Science site

6. Differentiation Checklist

For every joint lesson, prepare:

- One shared hook / investigation
- Grade 3 success criteria tied to their learner outcome
- Grade 4 success criteria tied to their learner outcome
- Two exit products (e.g. Gr 3 labelled diagram; Gr 4 evidence explanation)
- FNMI / local Alberta connection where curriculum names it
- Scientific Methods skill + optional CS extension

7. Quick Lesson Pattern

Step	Time	What Happens
Hook	5–10 min	Video or object demo for both
Shared investigation	20–25 min	Same materials
Split processing	15–20 min	Grade-specific notebook / centre
Share-out	5–10 min	Mixed pairs teach one idea
Assessment	Exit	Sticky-note ticket with grade-coded question

Sources

- Alberta Education Science curriculum via new LearnAlberta
- Alberta.ca Science snapshot and K–6 Science fact sheet
- ULeithbridge Curriculum Lab; APLC CPAR; Science Curriculum Wayfinder
- NASA, PBS LearningMedia, Crash Course Kids

Presentable planning document for classroom use. Adapt timelines to your school calendar.